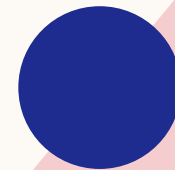


LAB 04
CASCADED ADDERS

CONTENTS

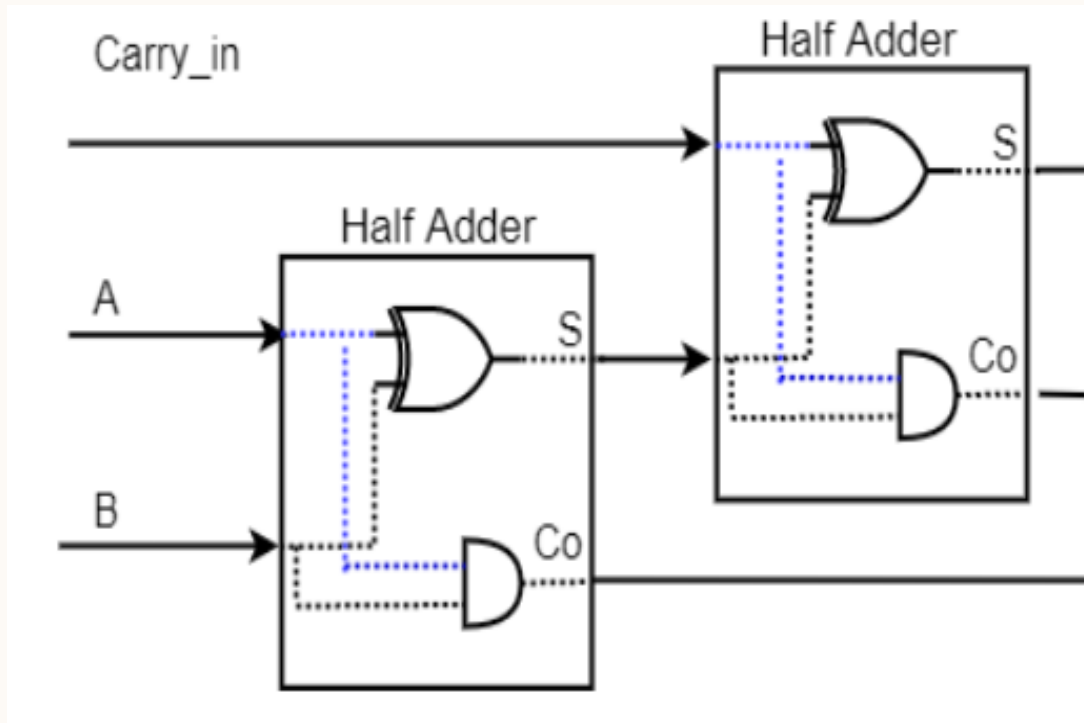
- Full Adder
- Cascade
- Cascading FAs (Ripple Carry Adder)
- Digital Display



FULL ADDER

3

- A combinational circuit which performs the arithmetic sum of three input bits is called **full adder**.
- The three inputs include **2 significant bits** and a **previous carry**.
- It can be implemented with **2 half-adders** and **1 OR gate**.

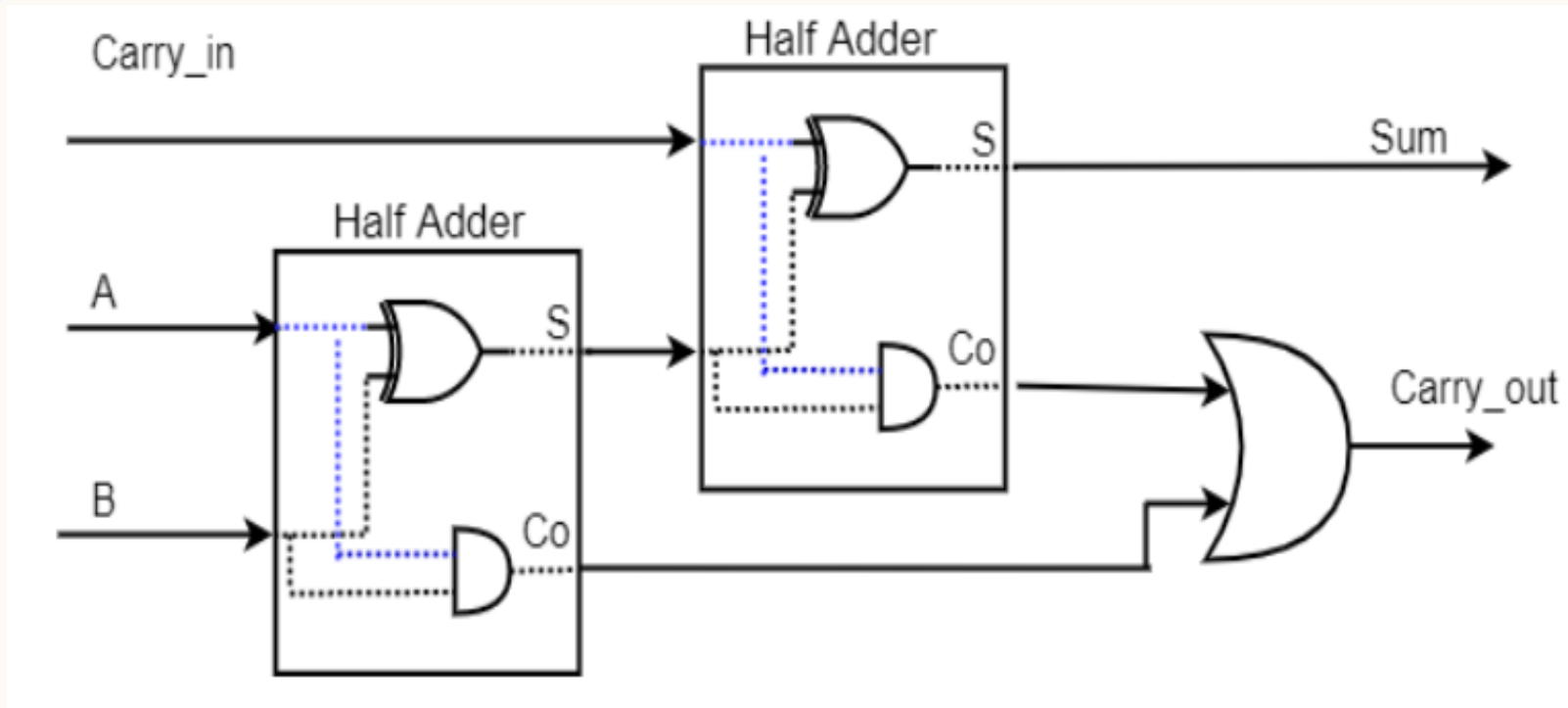


2 half adders

$$Sum = A \oplus B \oplus Cin$$

$$Carry = (A \oplus B) \cdot Cin + A \cdot B$$

FULL ADDER



Full adder using half adders

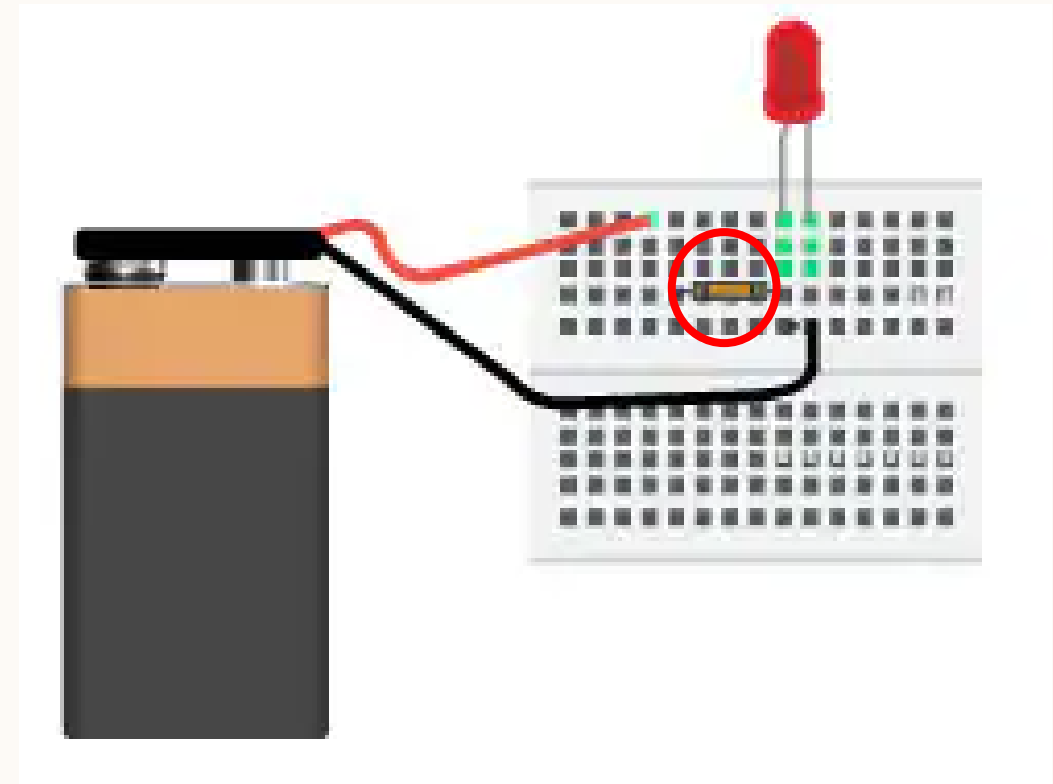
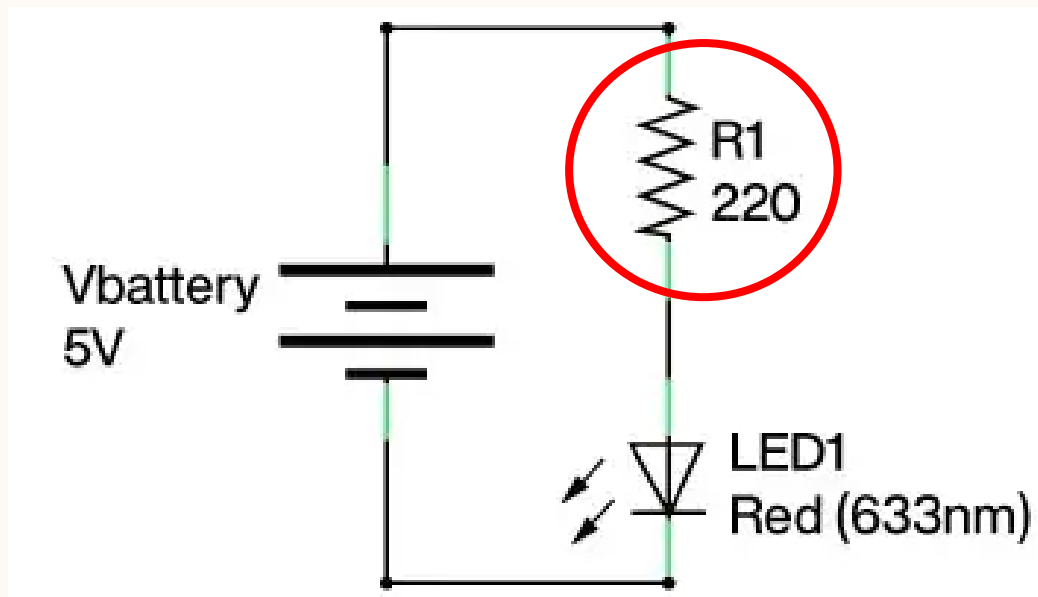
$$Sum = A \oplus B \oplus Cin$$

$$Carry = (A \oplus B) \cdot Cin + A \cdot B$$

In Task (a), connect output Sum & Carry_out to LEDs with current limiting resistor.

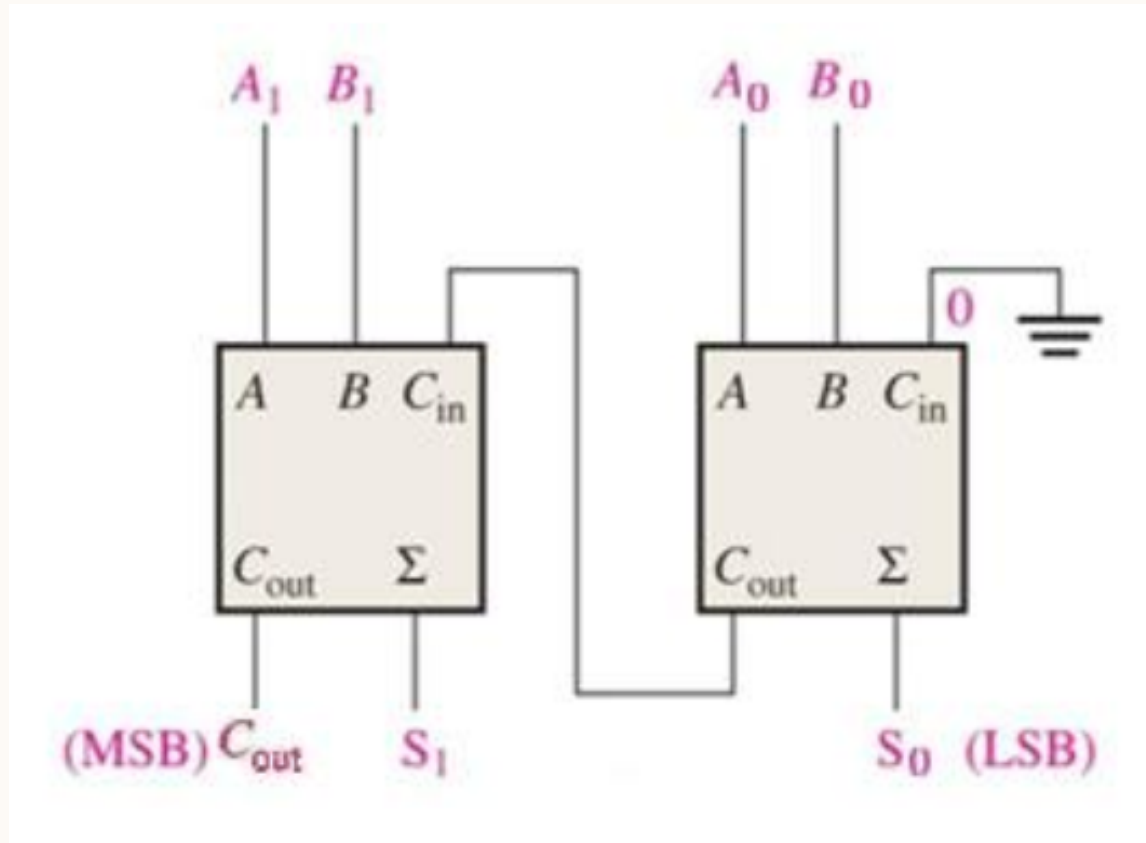
CURRENT LIMITING RESISTOR

- A current-limiting resistor is essential to prevent damage to an electric component (IC, LED) by controlling the electric current that flows through it, ensuring it operates at a safe, intended level.



CASCADING FULL ADDERS

- In engineering cascading represents several similar stages with each processing the output from the previous stage.
- Multiple full adders can be cascaded to add N -bit numbers.
- 2 full adders can be cascaded to add two 2-bit numbers.



$$S_0 = A_0 \oplus B_0 \oplus C_{in}$$

$$C_0 = (A_0 \oplus B_0) \cdot C_{in} + A_0 \cdot B_0$$

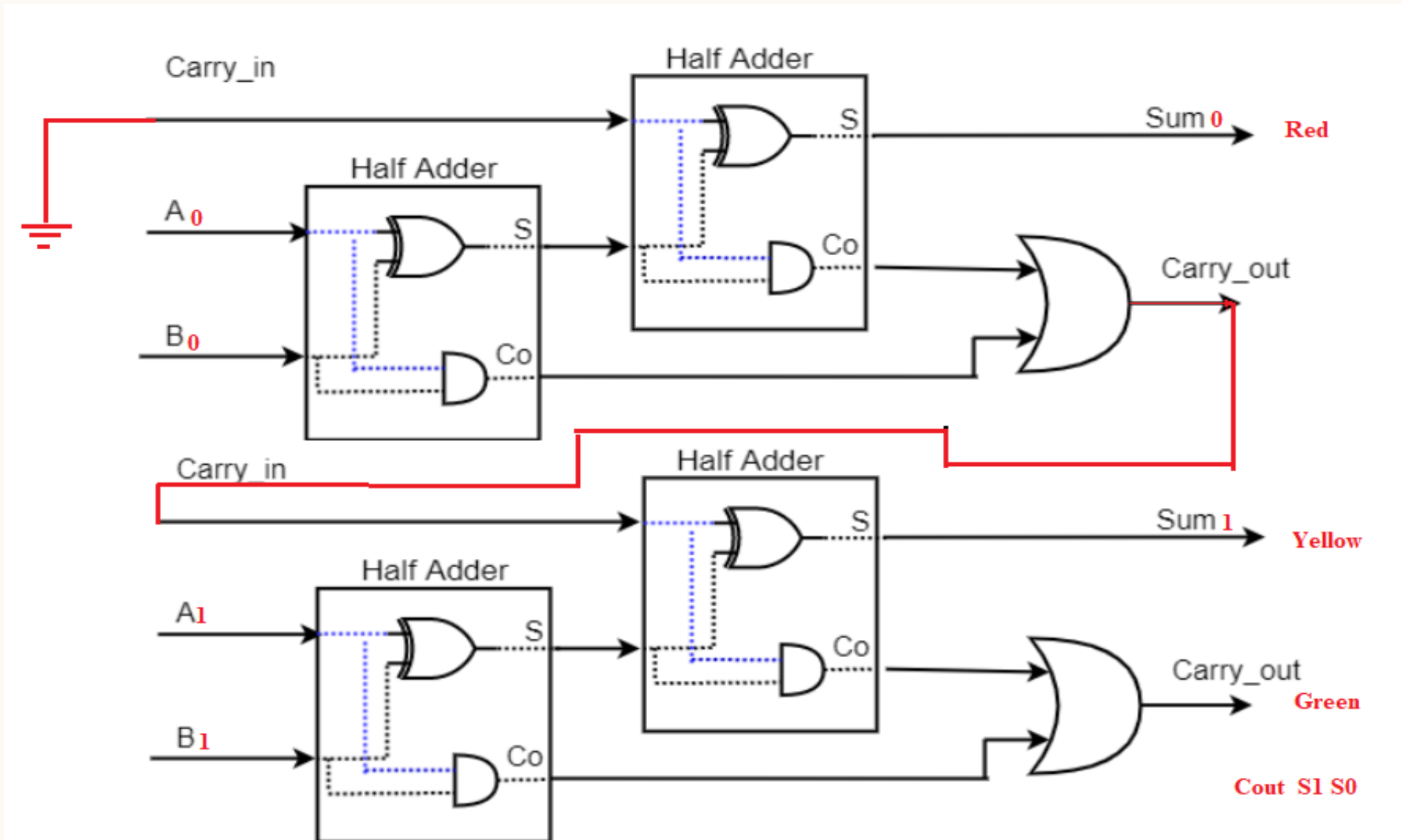
$$S_1 = A_1 \oplus B_1 \oplus C_0$$

$$C_1 = (A_1 \oplus B_1) \cdot C_0 + A_1 \cdot B_1$$

CASCADING 2 FULL ADDERS

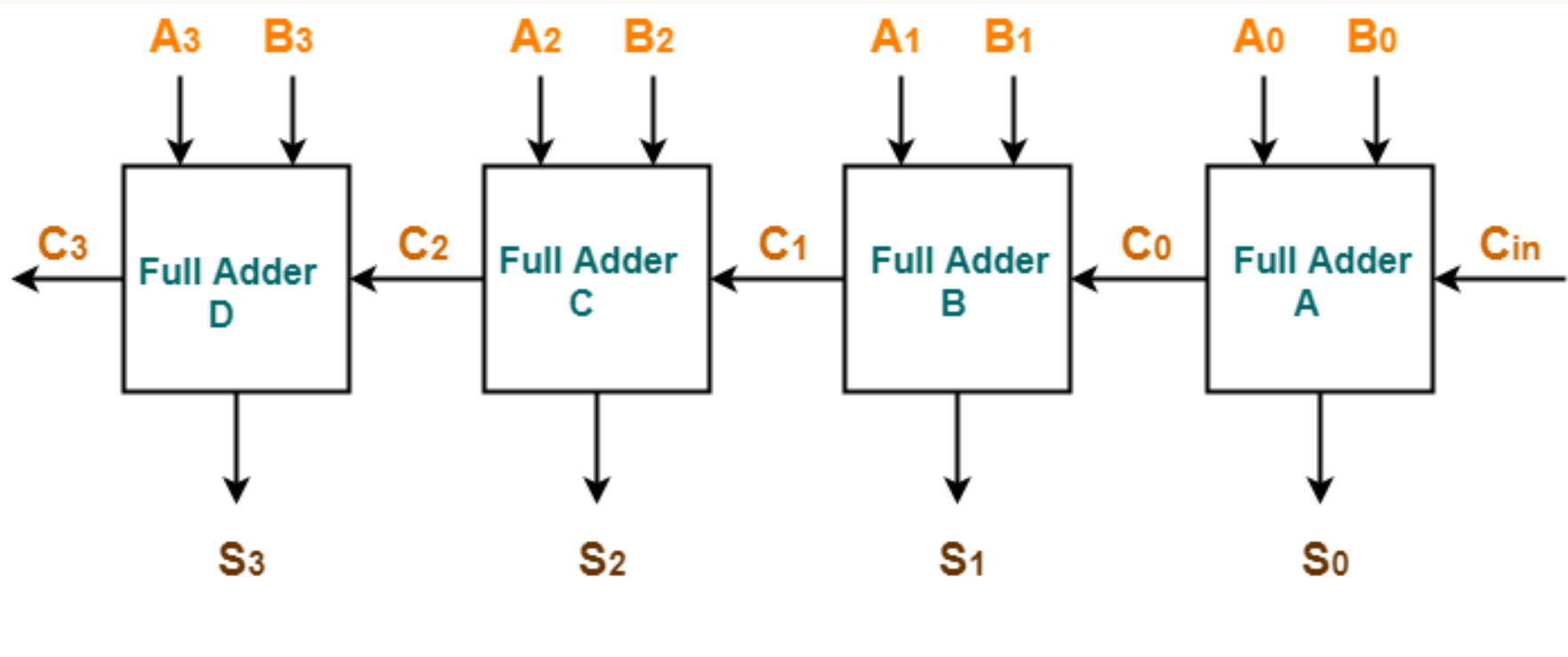
7

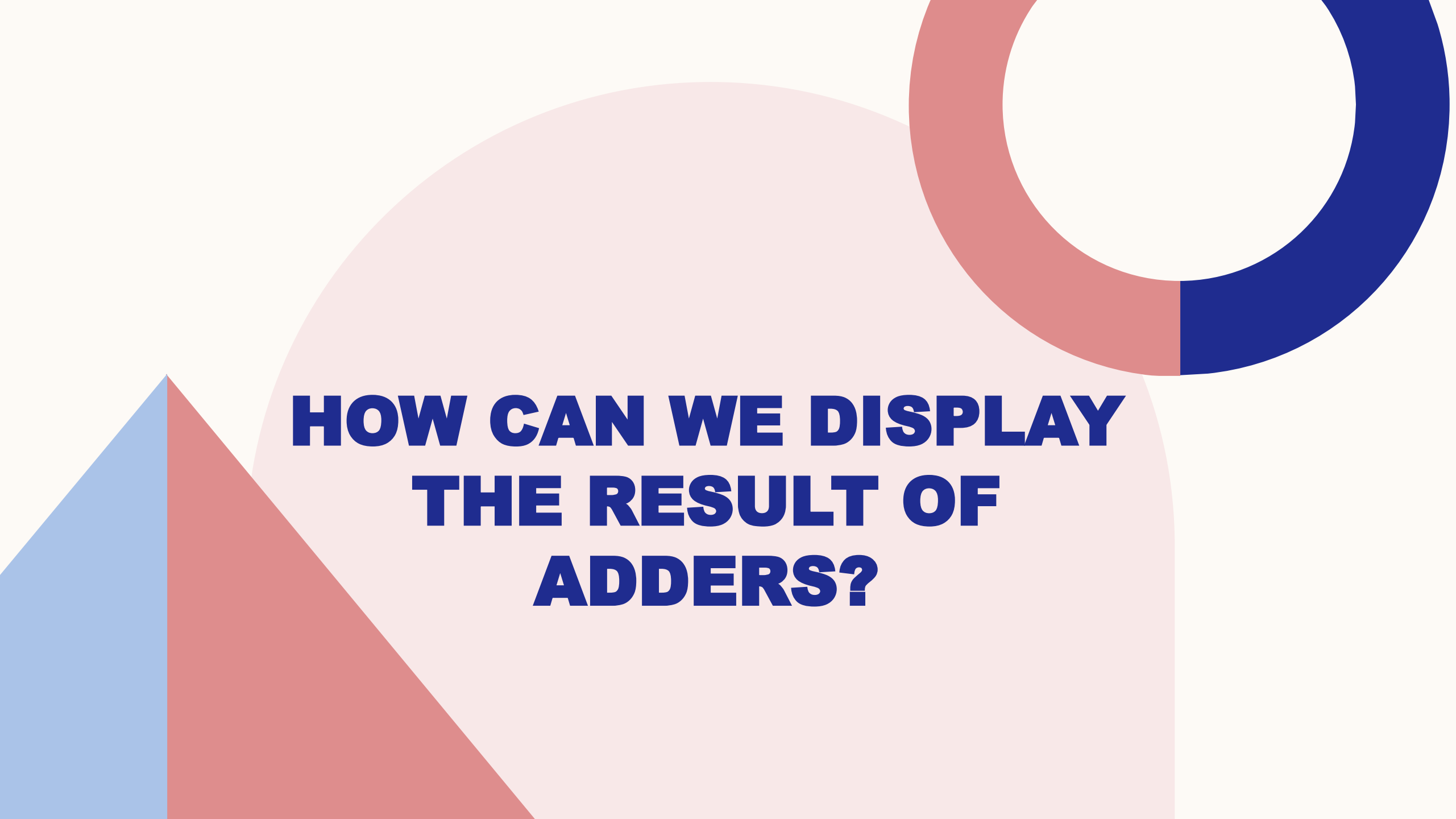
How many bits are there in the number that is being added (A&B) using 2 Cascaded FAs?



CASCADING FULL ADDERS

How many full adders are needed to add two 4 bit numbers?

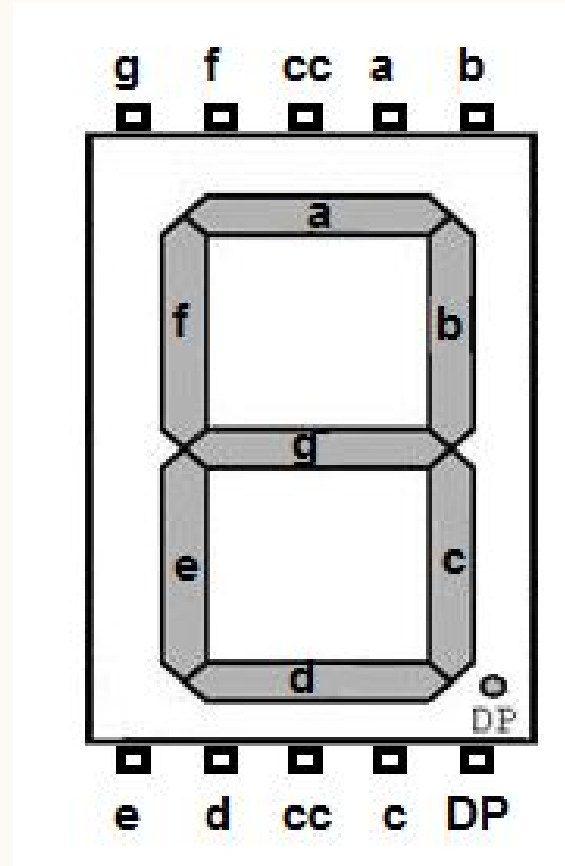




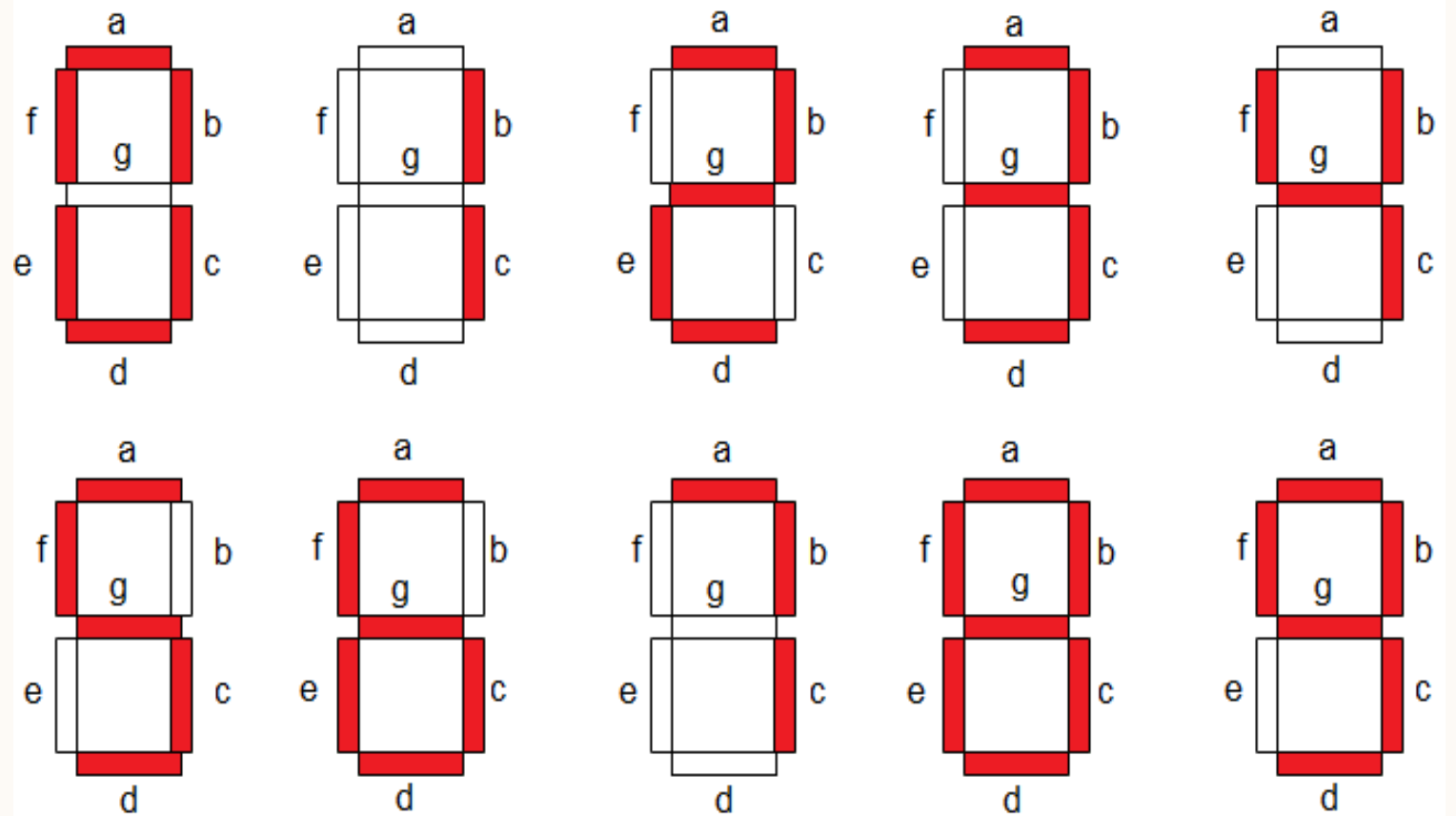
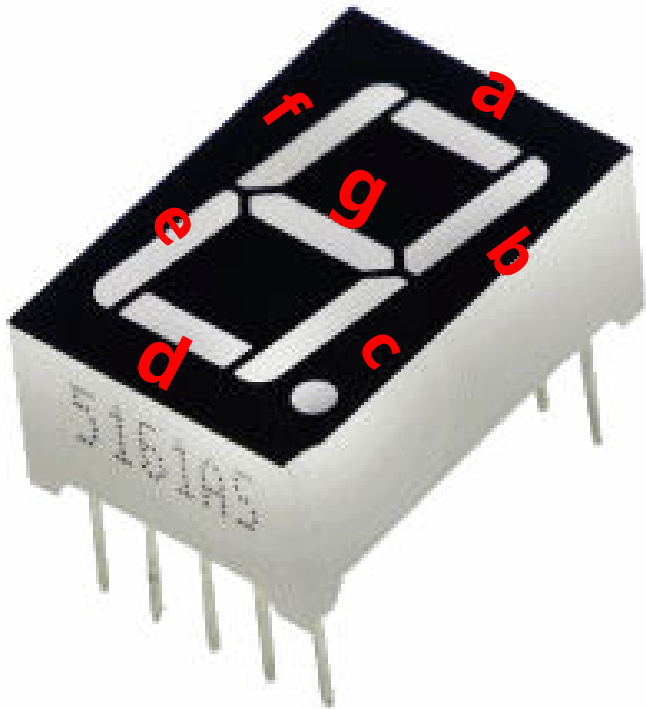
**HOW CAN WE DISPLAY
THE RESULT OF
ADDERS?**

SEVEN SEGMENT DISPLAY

- A 7-segment display is used in electronic display devices for decimal numbers from 0 to 9; and hexadecimal numbers from 0 – F.

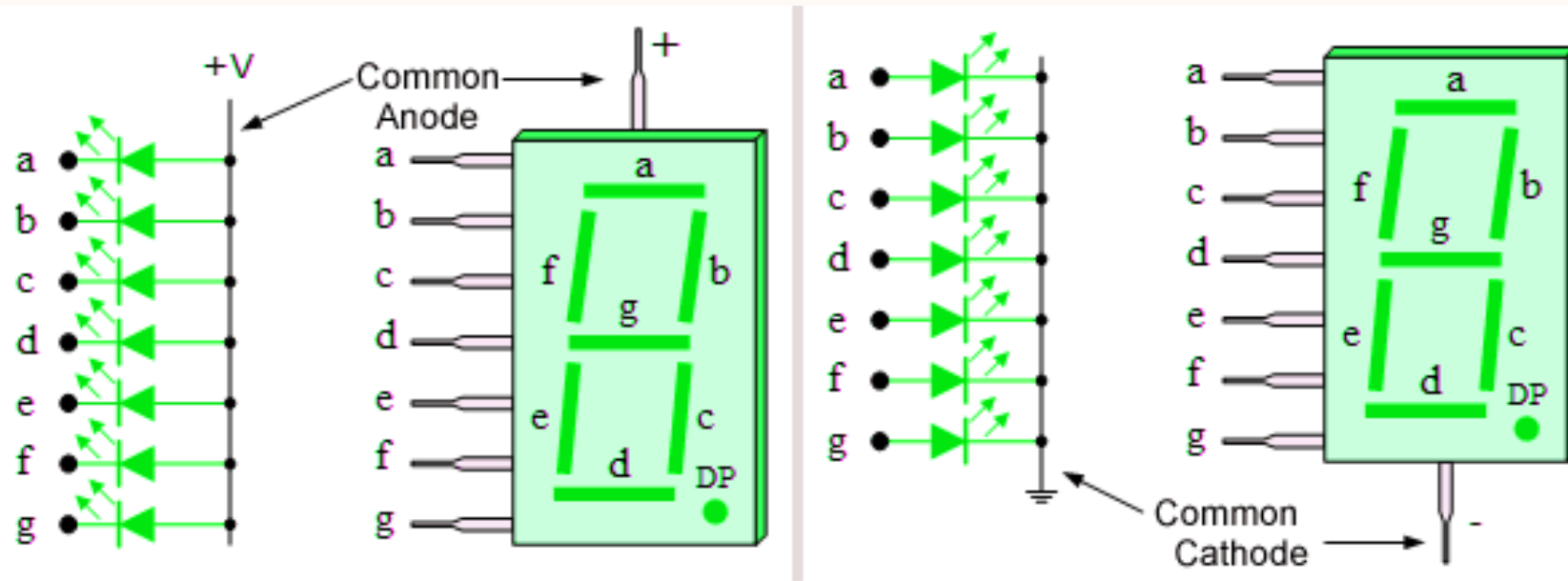


SEVEN SEGMENT DISPLAY

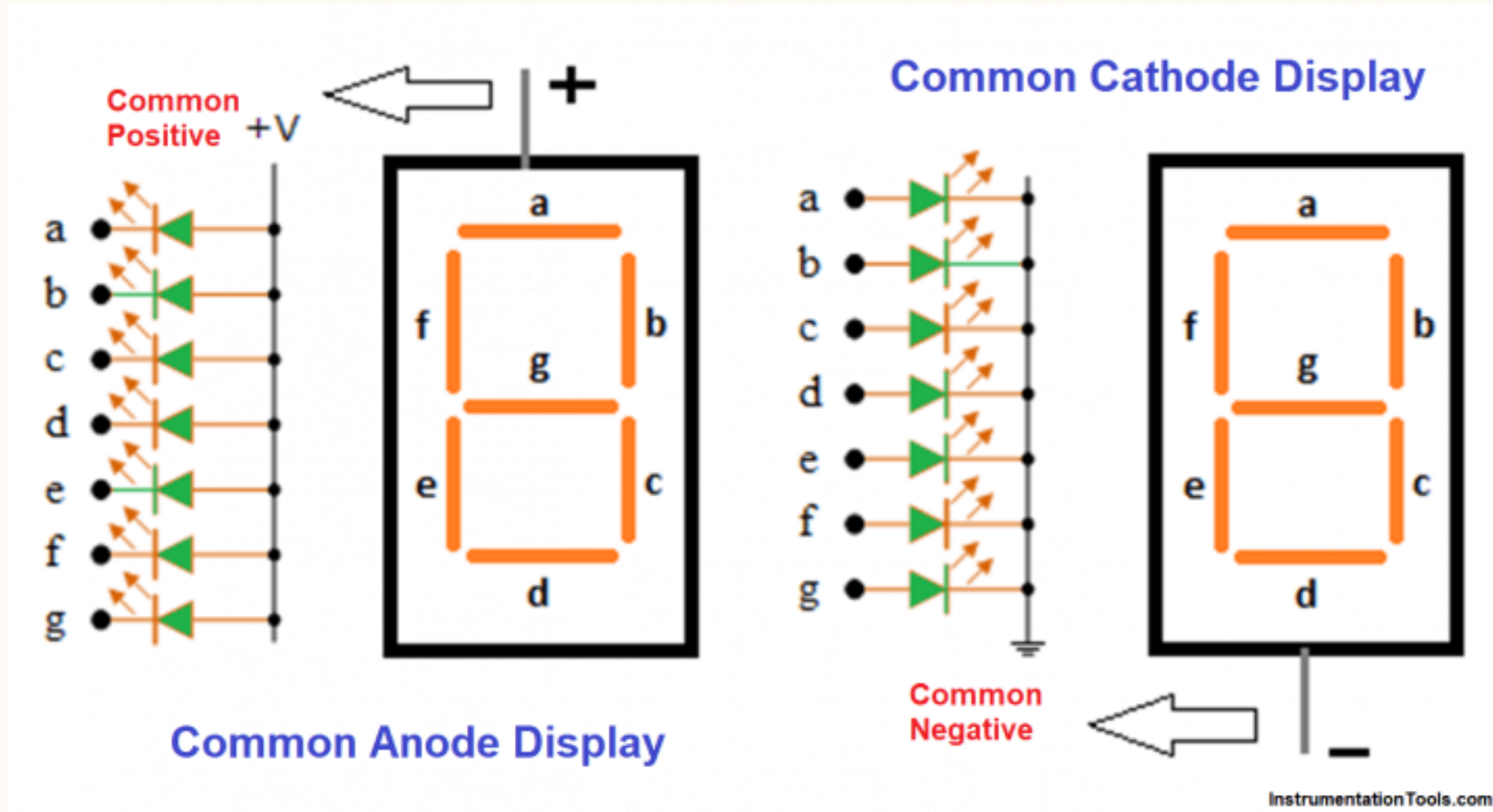


SEVEN SEGMENT DISPLAY

- LEDs in any 7-segment display are arranged in **Common Anode (CA) mode – Common Positive** or **Common Cathode (CC) mode – Common Negative**.
- In CA, the positive of every LED is connected and brought out as VCC.
- In CC, the negative terminals of every LED is connected and brought out as GND.



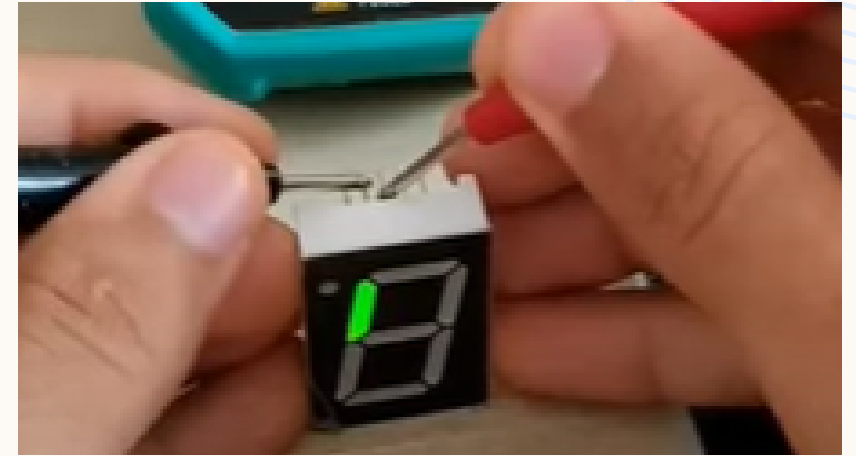
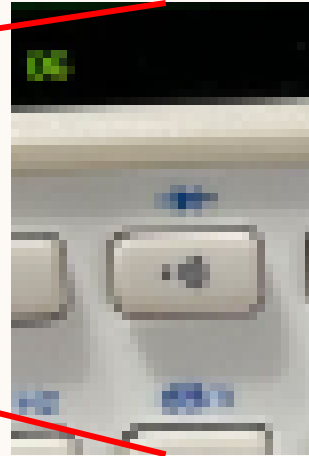
TEST YOUR 7-SEGMENT



TEST EACH SEGMENT OF 7-SEGMENT DISPLAY BY USING DMM IN DIODE MODE



Step 1: Put the DMM in diode mode (shift + 4th button from upper left).



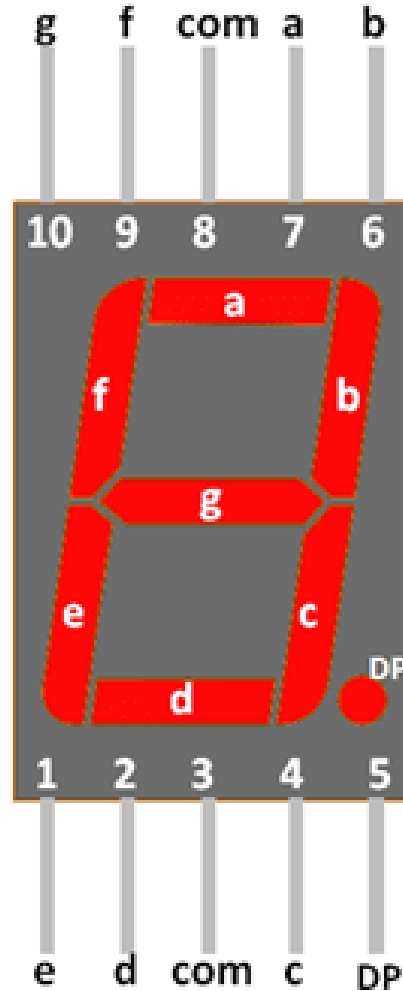
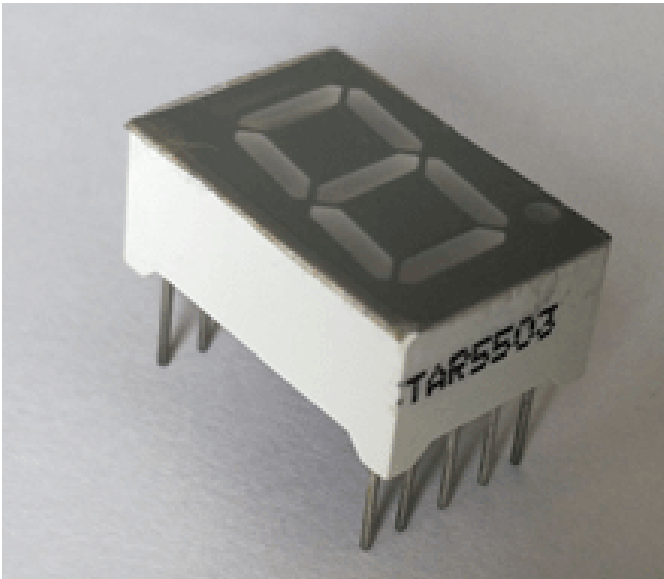
Step 2: Connect **Positive probe (red)** of DMM to Vcc pin of 7-segment display (Check Figure 4.4 for pin configuration of 7-segment display).

Step 3: Connect **Negative probe (black)** of DMM to the segment's pin of 7-segment display (Check Figure 4.4 for pin configuration of 7-segment display).

If LED of the segment lits up, the segment works.
All 7 segments must work for the 7-segment display to work properly.

❖ **Tutorial is available on LMS CANVAS**

SEVEN SEGMENT DISPLAY



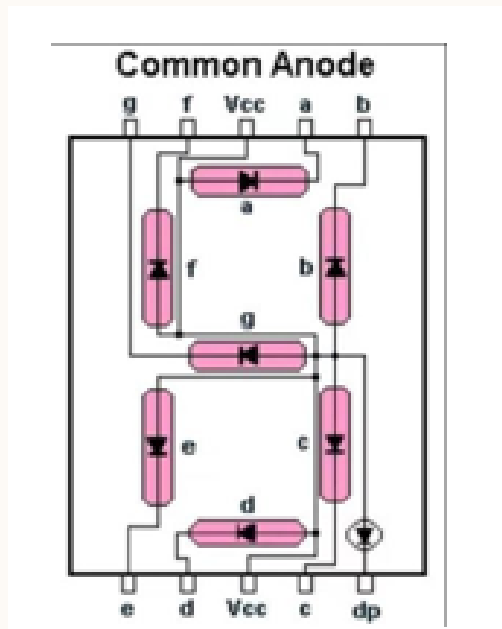
Connect Vcc (as we have CA 7-segment display) to **any one** of the com pin!

DP pin won't be connected to anything for this lab.

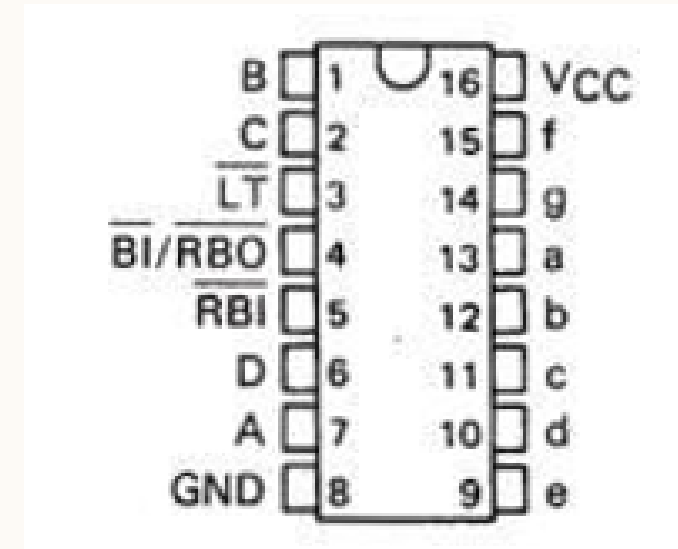
IC 7447 – 7-SEGMENT DISPLAY

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- **BCD (Binary Coded Decimal)** is a way of writing decimal numbers (0–9) in binary form.
- Each decimal digit is represented by **4 bits** (because 4 bits can count from 0000 to 1001 = 0 to 9).
- Example:
 - Decimal 5 → BCD = 0101
 - Decimal 9 → BCD = 1001



Pinout of 7-Segment display

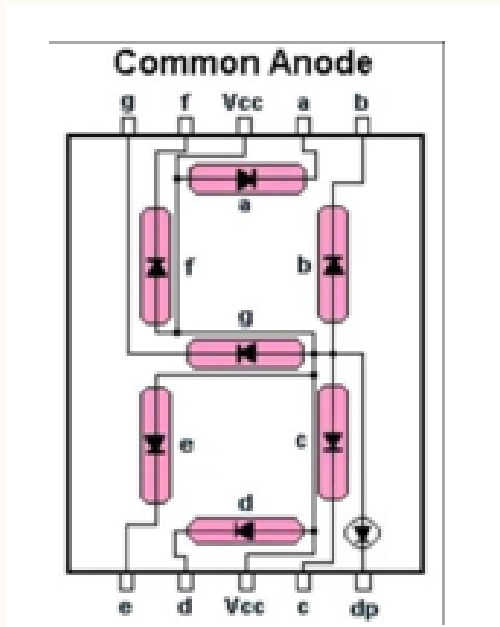


Pinout of 7447 IC

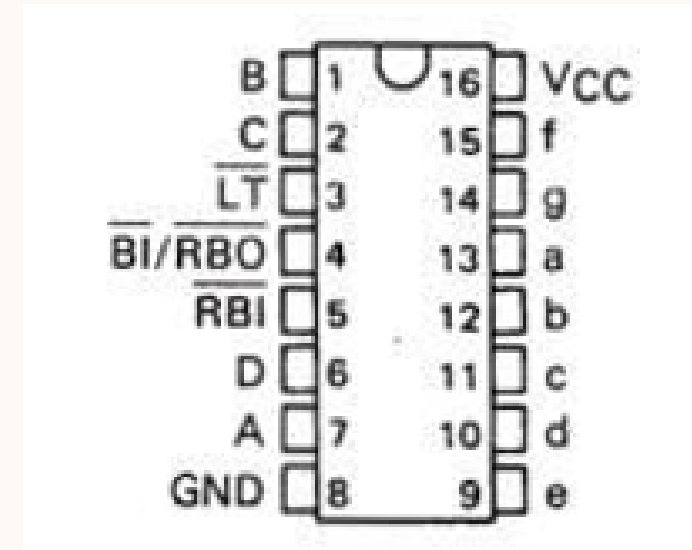
IC 7447 – 7-SEGMENT DISPLAY

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- Binary Coded Decimal to Seven Segment display.
 - Outputs are active low.
 - Sequence of input is DCBA.
 - You can also use 1k resistor as current limiting resistor.
- **DCBA:**
 - D:** The most significant bit (MSB).
 - C:** The next significant bit.
 - B:** The next significant bit.
 - A:** The least significant bit (LSB).
 - For two bits input: A₀ LSB – is connected to A. A₁ MSB – is connected to B. C & D will be grounded.



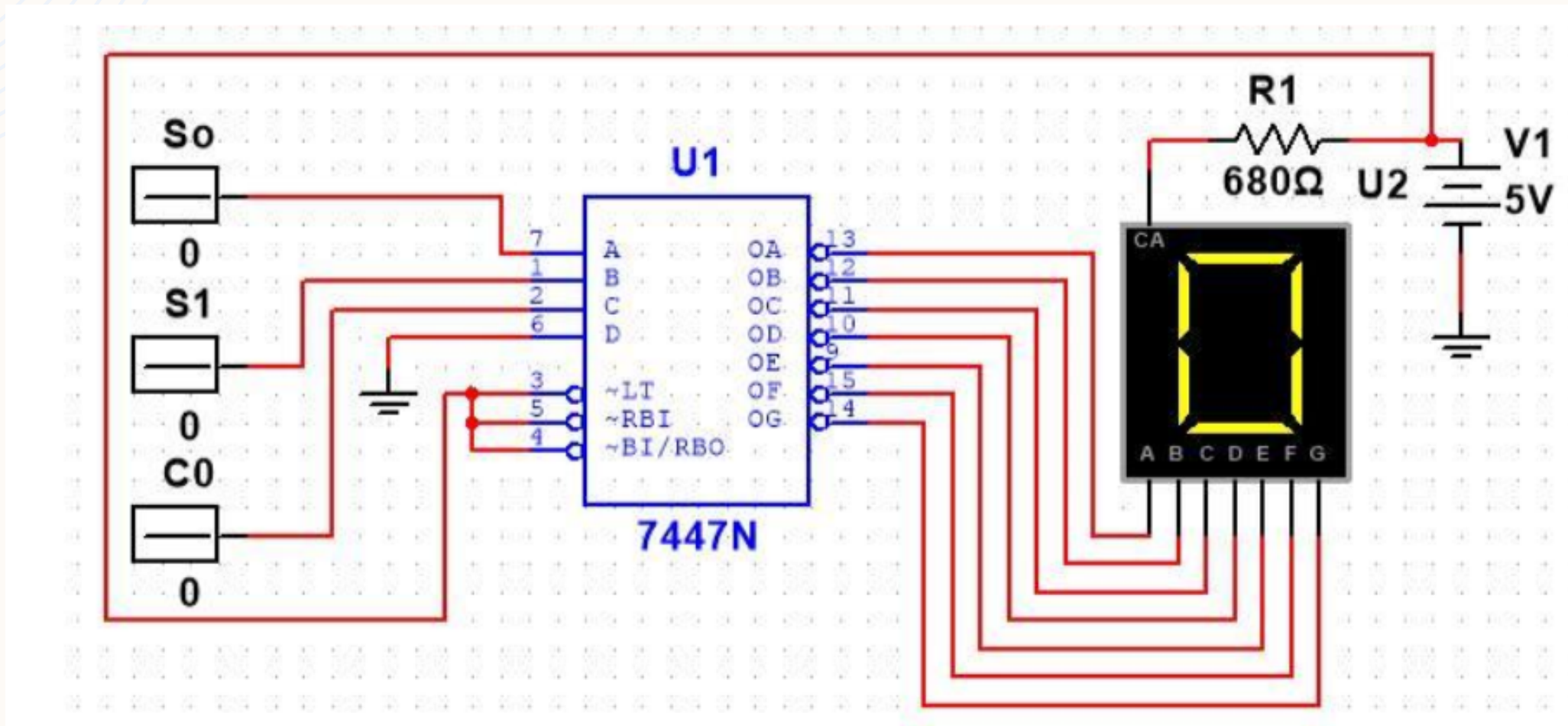
Pinout of 7-Segment display



Pinout of 7447 IC

TASK C

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Once the results on your 7-segment display are verified using slide switches, remove the slide switches from the circuit, and connect the output of 2-bit adder (Task b) to the 7447 IC inputs. **Get it checked by RA.**

TROUBLESHOOTING – CIRCUIT LAYOUT LR1

1. Check VCC and GND has been given to IC, use Multimeter to check the voltages on pin 7 and pin 14.
2. Check LED's polarity and gates of IC.
3. Check wires are inserted fully into the board.
4. Check power rails connection.
! There is a disconnection in the rail in center.
5. Double check IC number.

IN-LAB TASKS

- 0) Lecture
 - a) Full Adder – Maximum 30 minutes (includes troubleshooting)
 - b) **Group of 4: Combine FAs.** - 30 minutes (includes troubleshooting)
 - c) 7 segment testing (CA or CC) – 5 minutes use DMM
 - d) BCD to 7-segment Display – 45 minutes (Construct and test by giving some input e.g.: 100 (4))
 - e) Then join FA to 7447 – check it by RA
- ❖ Use two different color LEDs for **Carry** and **Sum** bit indication in Task a & b.